

Course Information Sheet

Course code: CL 24303
Course title: Chemical Reaction Engineering-II
Pre-requisite(s): Chemical Reaction Engineering-I (CL 24213)
Co- requisite(s):
Credits: L: 3 T: 1 P: 0
Class schedule per week: 4
Class: B. Tech
Semester / Level: V / Third
Branch: Chemical Engineering
Name of Teacher:

Course Objectives:

This course enables the students to:

1.	Explain the basic concepts of heterogeneous catalysis.
2.	Calculate and explain the physical properties of the catalysts.
3.	Development the kinetics of heterogeneous reactions
4.	Describe the reactors for various reactions.
5.	Develop critical and creative thinking skills related to reaction engineering.

Course Outcomes:

After the completion of this course, students will be able to:

CO24303.1	Describe the heterogeneous reactions and evaluate the physical properties of the catalysts.
CO24303.2	Derive the kinetics for heterogeneous catalytic reaction.
CO24303.3	Explain the transport effects of reaction in the heterogeneous catalysis.
CO24303.4	Derive the kinetics for heterogeneous non-catalytic reaction.
CO24303.5	Describe the kinetics of multiphase and biochemical reaction systems.

SYLLABUS

MODULE	NO. OF LECTURE HOURS
Module 1: Introduction to heterogeneous reactions. Classification of heterogeneous reactions. Concept of intrinsic and global rate of reaction. Components of the Catalyst formulation. Catalyst preparation. Determination of physical properties (BET surface area, pore volume, and pore size distribution) of heterogeneous catalysts.	8
Module 2: Steps in a catalytic reaction: Adsorption isotherms, Surface reaction, Desorption. Rate limiting step. Mechanism and synthesizing a rate law.	8
Module 3: Kinetics of solid catalyzed gas phase reaction; isothermal and non-isothermal inter-and-intraphase effectiveness factors, catalytic gas-solid reactor design.	8
Module 4: Different types of non-catalytic solid-gas reactions; kinetic models for gas- solid non-catalytic reactions: Shrinking Core Model (SCM), Progressive conversion Model (PCM), Design of non-catalytic gas solid reactors.	8
Module 5: Gas liquid reactions: Enhancement factor in gas-liquid reactions, design of gas-liquid reactors. Gas-Liquid Reactions on Solid Catalyst, Suspended Solids Reactors: Bubbling fluidized bed (BFB), K-L Model for BFB.	8

Textbooks:

1. J. J. Carberry, Chemical and Catalytic Reaction Engineering, Dover Books on Chemistry, 2001.
2. Chemical Engineering Kinetics, J. M. Smith.
3. Elements of Chemical Reaction Engineering, Fogler H. S., Prentice Hall, 2001.
4. O. Levenspiel, Chemical Reaction Engineering, 3rd Edn, Wiley & Sons (1999).

Reference books:

1. Chemical Reactor Analysis and Design Gilbert F. Froment, Kenneth B. Bischoff, Juray De Wilde, John Wiley & Sons, Incorporated, 2010

Gaps in the syllabus (to meet Industry/Profession requirements)**POs met through Gaps in the Syllabus****Topics beyond syllabus/Advanced topics/Design****POs met through Topics beyond syllabus/Advanced topics/Design**

Course Outcome (CO) Attainment Assessment tools & Evaluation procedure

Direct Assessment

Assessment Tool	% Contribution during CO Assessment
First Quiz	10
Mid Semester Examination	25
Second quiz/seminar/assignment/others	10
Teacher's Assessment	5
End Semester Examination	50

Indirect Assessment –

1. Student Feedback on Course Outcome

Mapping of Course Outcomes onto Program Outcomes

Course Outcome	Program Outcomes (POs)												Program Specific Outcomes (PSOs)		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
CO24303.1	3	3	3	2	2	2	2	1	1	2	1	3	3	3	3
CO24303.2	3	2	1	2	3	1	1	1	1	2	1	3	3	3	3
CO24303.3	3	1	1	1	1	1	1	1	1	1	1	1	3	3	3
CO24303.4	3	3	2	2	3	1	2	1	1	2	1	3	3	3	3
CO24303.5	3	3	1	2	3	2	2	1	1	2	1	3	3	3	3

Correlation Levels 1, 2 or 3 as defined below:

- 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Mapping Between COs and Course Delivery (CD) methods:

CD Code	Course Delivery methods	Course Outcome	Course Delivery Method Used
CD1	Lecture by use of boards/LCD projectors/OHP projectors	CO1	CD1, CD2, CD8
CD2	Tutorials/Assignments	CO2	CD1, CD2, CD8
CD3	Seminars	CO3	CD1, CD2, CD8
CD4	Mini projects/Projects	CO4	CD1, CD2, CD8
CD5	Laboratory experiments/teaching aids	CO5	CD1, CD2, CD8
CD6	Industrial/guest lectures		
CD7	Industrial visits/in-plant training		
CD8	Self- learning such as use of NPTEL materials and internets		
CD9	Simulation		